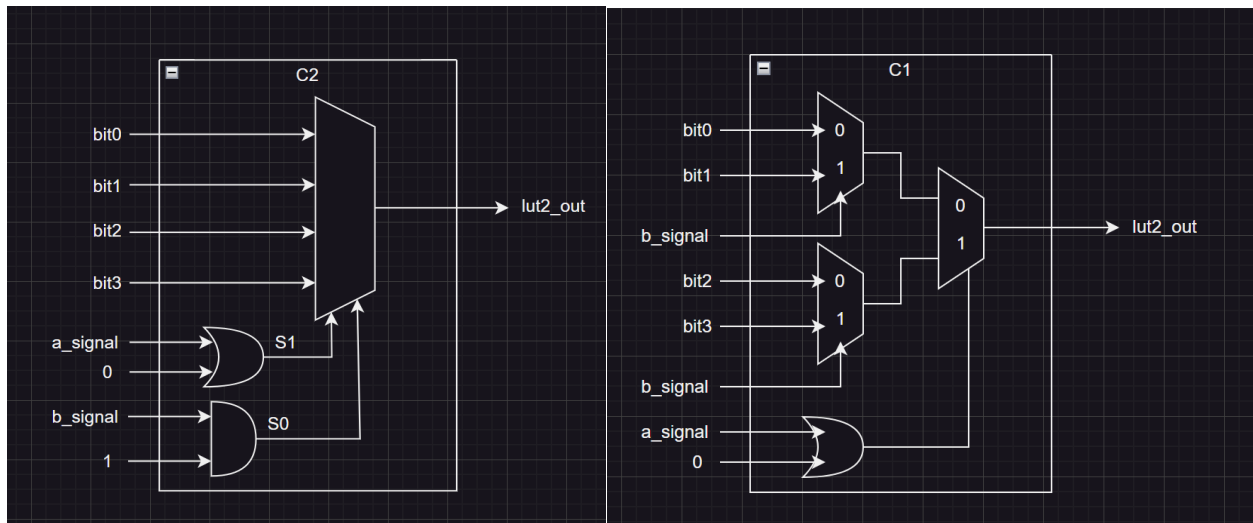


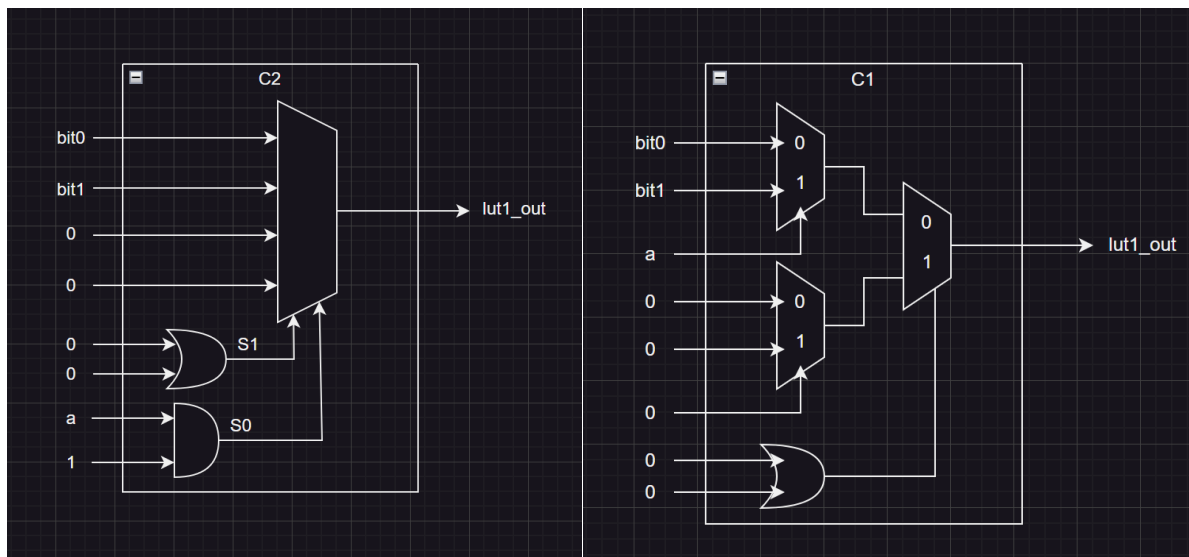
Mapping LUT2 cells:

Diagrams below provide the implementation of LUT2 using ACTEL C1 and C2 components.



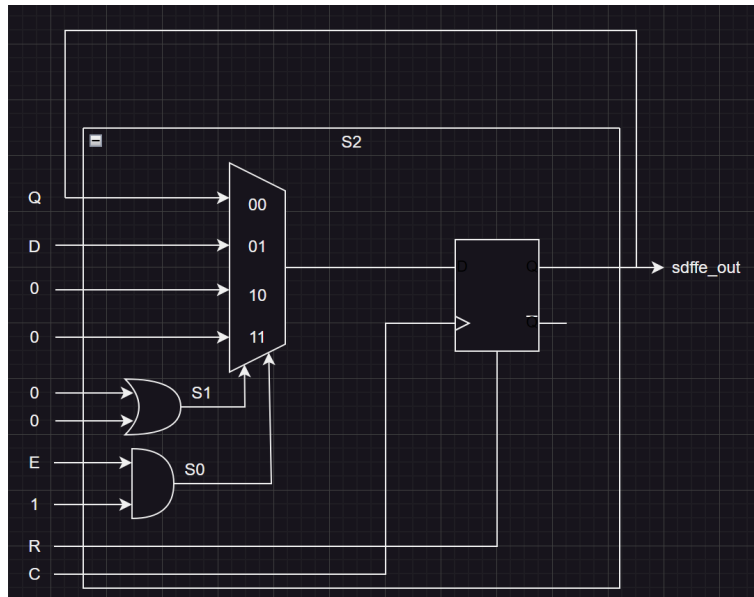
Mapping LUT1 cells:

Diagrams below provide the implementation of LUT1 using ACTEL C1 and C2 components.



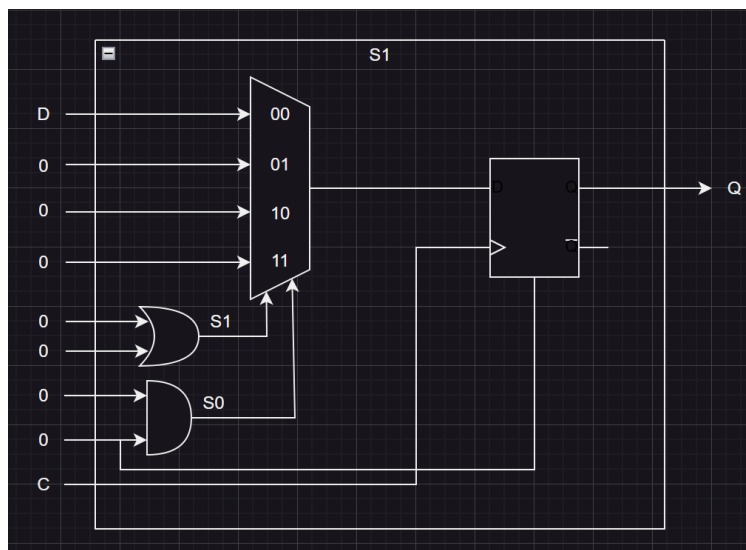
Mapping SDFFE cell:

Diagram below provides the implementation of SDFFE using ACTEL S2 cell.



Mapping DFF cell:

Diagram below provides the implementation of DFF using ACTEL S1 cell.



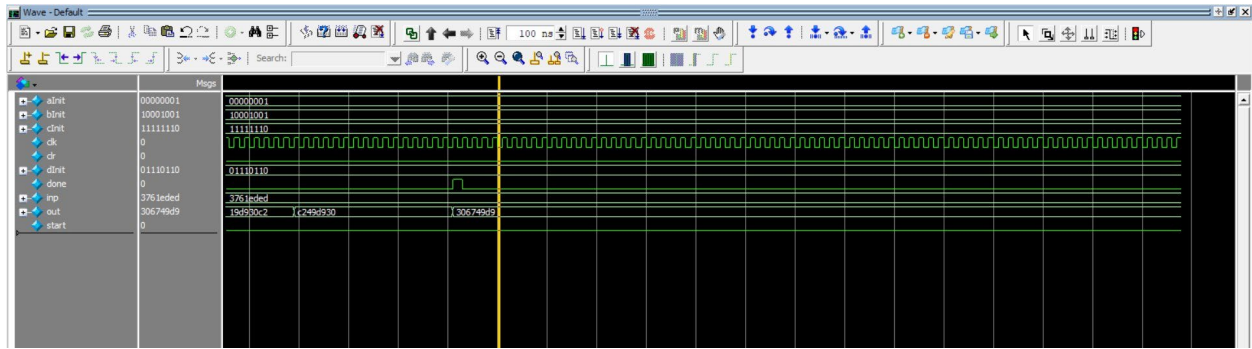
Results:

After running YOSYS on our original code modified to accommodate changes in assignment 3 (adding multiplier and the length of registers), we obtained a the HashGenerator_lut2.v file which contains the program logic but only based on units of LUT.

Then we use this file as an input for our python mapper code which in turn generates the file mapped_HashGenerator_lut2.v that contains the program logic, but this time only based on ACTEL units (C1, C2, S1, S2).

Then we proceeded to simulate the resulting mapped Verilog file. This was the hardest part of this assignment. It requires us to understand several synthesis errors and fix them without breaking the logic of our code.

After finally fixing all errors, we were able to simulate the Verilog file, and the following figure illustrates the waveform of the simulation.



The python mapping code also provided us with several statistical information which are shown in the following figure:

```
Found 0 LUT1 and 717 LUT2 instances (out of 717 $lut)
Total number of c2: 717
Total number of S1: 27
Total number of S2: 25
Created mapped design: mapped_HashGenerator_lut2.v
Created modules file: modules.v
PS D:\5th_semester\CAD\Computer-Aided-Designs-CA4\code>
```

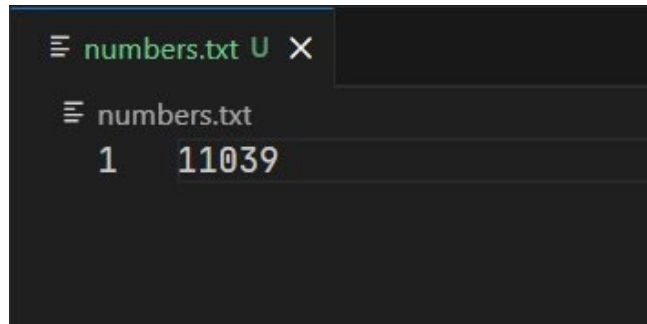
From the above figure we can tell that YOSYS used 717 LUT2 components and then after mapping them to ACTEL components we achieved a total number of 717 C2, 27 S1 and 25 S2 components.

Eventually we can run the counter program from the last assignment to calculate the total number of logical gates used in the mapped code and compare it with our own implementation.

```
numbers.txt
1 2982
```

Recall from the last assignment that we achieved this number shown in the above figure as the total number of logical gates used in the process of mapping our design to ACTEL components.

Running the counter program for the mapped code we get the following result:



```
≡ numbers.txt U X
≡ numbers.txt
1 11039
```

As can be observed from the above figure, the number of gates increased significantly.

This shows that despite the fact that automative synthesis tools have been developed to optimize the designs and to reach as few components as possible, our manual design and mapping achieved a significant lower number of gates.